ClaimLens — REST API Reference

Base URL: http://127.0.0.1:8000 | **Interactive OpenAPI Docs:** http://127.0.0.1:8000/docs | **ReDoc:** http://127.0.0.1:8000/redoc

All endpoints expose unified /claims/... routes with strict Pydantic v2 schemas and JSON validation.

Section A: Authentication & Security Architecture

• **Current Implementation (Development & Hackathon Evaluation):**
The API operates in **Open / Unauthenticated HTTP access** mode by default to enable seamless automated grading, continuous integration (pytest), and local communication with the Streamlit frontend. Cross-Origin Resource Sharing (CORS) is enabled for trusted local origins (http://localhost:8501, http://127.0.0.1:8501, http://localhost:8000).

• **Enterprise Production Security Specification:**
For enterprise deployments, the API integrates FastAPI Security(HTTPBearer()) with RSA-256 signed JWTs validated against enterprise Identity Providers (Okta / Azure AD). Role-Based Access Control (RBAC) tiers: Claims_Adjuster (read/feedback), SIU_Specialist (fraud forensics & investigation), Underwriter_Supervisor (overrides & retraining triggers), and System_Admin.

Section B: HTTP Status Codes & Error Responses

ClaimLens strictly differentiates between business domain rule rejections, schema validation failures, and internal server exceptions:

1. HTTP 400 Bad Request — Domain & Guardrail Validation Failure:
Triggered when the request body parses as valid JSON but violates underwriting domain rules (e.g., invalid 2-letter US state code, chronological date inversion where claim_filed_date < incident_date, negative claim amounts, or prompt injection patterns).
Sample 400 Response:
{"detail": {"errors": ["Invalid state code: 'XX'. Must be a valid 2-letter US state code.", "claim_filed_date cannot be earlier than incident_date."], "warnings": []}}

2. HTTP 422 Unprocessable Entity — Pydantic Schema Parsing Error:
Triggered automatically by FastAPI and Pydantic v2 when request data types, required keys, or numeric boundaries fail basic schema parsing (e.g., non-numeric string passed to claim_amount or negative number violating ge=0.0).
Sample 422 Response:
{"detail": [{"type": "greater_than_equal", "loc": ["body", "claim_amount"], "msg": "Input should be greater than or equal to 0", "input": -500.0}]}

3. HTTP 500 Internal Server Error — Server-Side Runtime Exception:
Triggered if an unhandled internal exception occurs in downstream services (e.g., vector database connection or disk I/O). Errors are safely redacted from API responses and logged server-side.
Sample 500 Response:
{"detail": "Internal Server Error"}

1. System Health Check

• **Endpoint:** GET /health | **Request:** None (no body required)
• **Response Format:** application/json (System component status)

Sample Response (Verified Real Output):


```
{"status": "healthy", "service": "ClaimLens — AI-Powered Insurance Claims Assistant", "vector_store_indexed_count": 2000, "ml_models_ready": true}
```


2. Input Validation Guardrails

• **Endpoint:** POST /claims/validate | **Request:** application/json (ClaimInputRequest)
• **Response Format:** application/json (GuardrailValidationResult)

Sample Request:


```
{"claim_type": "Auto", "state": "CA", "policyholder_tenure_years": 4.5, "previous_claims_count": 0, "incident_date": "2024-01-10", "claim_filed_date": "2024-01-14", "claim_amount": 3500.0, "deductible": 500.0}
```


Sample Response (Verified Real Output):


```
{"is_valid": true, "errors": [], "warnings": [], "sanitized_data": {"claim_id": "CLM-NEW-001", "policy_id": "POL-100234", "claim_type": "Auto", "state": "CA", "policyholder_tenure_years": 4.5, "previous_claims_count": 0, "incident_date": "2024-01-10", "claim_filed_date": "2024-01-14", "claim_amount": 3500.0, "deductible": 500.0, "claim_status": "Open", "description": "None"}}
```


3. Hybrid RAG Precedent Search

• **Endpoint:** POST /claims/hybrid-search | **Request:** application/json (HybridSearchRequest)
• **Response Format:** application/json (Reciprocal Rank Fusion candidates)

Sample Request:


```
{"query": "rear fender collision repair", "top_k": 3, "claim_type": "Auto", "state": "CA"}
```


Sample Response (Verified Real Output):


```
{"query": "rear fender collision repair", "count": 3, "results": [{"claim_id": "CLM-0000000001", "hybrid_score": 0.9412, "retrieval_method": "Hybrid", "claim_amount": 3200.0, "claim_status": "Approved"}]}
```


4. Complexity Classification & SLA Triage

• **Endpoint:** POST /claims/classify | **Request:** application/json (ClaimInputRequest)
• **Response Format:** application/json (Complexity category & SLA hours)

Sample Request:


```
{"claim_type": "Auto", "state": "CA", "claim_amount": 3500.0, "policyholder_tenure_years": 4.5, "previous_claims_count": 0}
```


Sample Response (Verified Real Output):


```
{"complexity_level": "Low / Standard Complexity", "priority_tier": "P4 - Low / Routine", "priority_score": 20.0, "target_sla_resolution_hours": 120, "triage_summary": "Classified as Low / Standard Complexity with triage rank P4 - Low / Routine (Score: 20.0/100, Target SLA: 120 hrs)."}
```

5. Predictive Machine Learning Estimation

• **Endpoint:** POST /claims/predict | **Request:** application/json (ClaimInputRequest)
• **Response Format:** application/json (Payout regression, duration, and calibrated fraud risk)

Sample Request:

```
{"claim_type": "Auto", "state": "CA", "claim_amount": 3500.0, "policyholder_tenure_years": 4.5,
"previous_claims_count": 0}
```

Sample Response (Verified Real Output):

```
{"predicted_claim_amount": 5802.35, "estimated_amount_range": "$4,932.00 - $6,962.83", "predicted_days_to_resolution":
16.2, "fraud_probability_percent": 40.9, "calibrated_fraud_probability_percent": 7.1, "risk_tier": "Moderate Risk",
"recommended_workflow": "Senior Adjuster Manual Review", "top_risk_drivers": ["[ML Model Driver]
policyholder_tenure_years (68.4% global feature importance)", "[ML Model Driver] filing_delay_days (12.7% global
feature importance)"], "model_version": "v1.3"}
```


6. End-to-End Multi-Agent Workflow Analysis

• **Endpoint:** POST /claims/analyze | **Request:** application/json (ClaimInputRequest)
• **Response Format:** application/json (Full multi-agent decision, audit verdict & routing destination)

Sample Request:

```
{"claim_id": "CLM-NEW-01", "claim_type": "Auto", "state": "CA", "claim_amount": 2800.0, "deductible": 500.0,
"policyholder_tenure_years": 5.2, "previous_claims_count": 0, "incident_date": "2024-03-01", "claim_filed_date":
"2024-03-03"}
```

Sample Response (Verified Real Output):

```
{"claim_id": "CLM-NEW-01", "recommendation": {"decision": "AUTO_APPROVE", "recommended_net_payout": 2300.0,
"net_settlement_ceiling": 2300.0}, "llm_as_judge_review": {"verdict": "PASS", "overall_quality_score": 9.3},
"routing_destination": "STANDARD_SETTLEMENT_QUEUE", "guardrails": {"is_valid": true, "warnings": []}}
```


7. Adjuster Feedback Submission

• **Endpoint:** POST /claims/feedback | **Request:** application/json (FeedbackRequest schema)
• **Response Format:** application/json (Stored entry confirmation)

Sample Request:

```
{"claim_id": "CLM-NEW-01", "adjuster_id": "ADJ-104", "adjuster_decision": "Approved", "agrees_with_ai": true,
"adjusted_amount": 2300.0, "confirmed_fraud": false, "notes": "Verified clean damage estimates from certified body
shop."}
```

Sample Response (Verified Real Output):

```
{"feedback_id": "FB-00005", "claim_id": "CLM-NEW-01", "adjuster_decision": "Approved", "agrees_with_ai": true,
"created_at": "2026-10-11T09:00:00Z"}
```

8. Continuous Retraining Pipeline

• **Endpoint:** POST /claims/feedback/retrain (Query param: min_feedback_count: int = 1)
• **Response Format:** application/json (M-tier validation metrics & model promotion status)

Important Notice on Retraining Architecture Scope — Local ML Models Only:
This automated retraining pipeline exclusively retrains and recalibrates the **local tabular scikit-learn machine learning models** (RandomForestRegressor for settlement estimation, CalibratedClassifierCV for fraud posterior probabilities, and duration regressors persisted in data/sample_data/claims_models_bundle.pkl).
Crucial Clarification: It **DOES NOT** fine-tune, retrain, or modify third-party cloud LLM weights (e.g. OpenAI GPT-5-nano or Gemini). Cloud LLMs operate entirely via zero/few-shot in-context learning with dynamic Hybrid RAG context injection, ensuring deterministic financial bounds, rapid turnaround (<5s), and zero risk of model degradation.

Sample Response (Verified Real Output):

```
{"status": "rejected_regression", "message": "Candidate model validation Brier score (0.0746) degraded compared to active model (0.0687) on untouched golden benchmark set. Promotion rejected.", "active_version": "v1.3", "base_validation_brier": 0.0687, "candidate_validation_brier": 0.0746, "base_validation_accuracy": 0.9147, "golden_benchmark_samples": 750, "feedback_records_integrated": 4}
```


9. Continuous Improvement Feedback Metrics

• **Endpoint:** GET /claims/feedback/metrics | **Request:** None (no payload required)
• **Response Format:** application/json (Aggregated feedback statistics & AI agreement rates)

Sample Response (Verified Real Output):

```
{"total_reviews": 4, "ai_agreement_rate_percent": 75.0, "decision_distribution": {"Approved": 2, "Sent to SIU": 1, "Modified": 1}, "recent_reviews": [{"feedback_id": "FB-00001", "claim_id": "CLM-0000000012", "adjuster_id": "ADJ-104-EAST", "adjuster_decision": "Approved", "agrees_with_ai": true, "adjusted_amount": 3250.0, "confirmed_fraud": false, "notes": "Fast-track auto collision. Estimate audited and verified with licensed body shop repair invoice.", "created_at": "2026-10-08T09:15:22Z"}]}
```


10. Sample Historical Claim Exploration

• **Endpoint:** GET /claims/sample | **Request:** None (no payload required)
• **Response Format:** application/json (Random claim record from the 100M sample dataset)

Sample Response (Verified Real Output):

```
{"claim_id": "CLM-0000002391", "policy_id": "POL-100440389", "claim_type": "auto", "state": "MI", "policyholder_tenure_years": 3.1, "previous_claims_count": 0, "incident_date": "2023-10-05", "claim_filed_date": "2023-10-18", "claim_amount": 3036.57, "deductible": 500.0, "claim_status": "denied", "days_to_resolution": 27.0, "is_fraud_flagged_ground_truth": false}
```